

MH1100 Calculus I

Handout 6 - Derivatives I

We will be focusing on the concept of the **derivative** of a function.

- The definition of derivative at a point a .
- The derivative function.
- How are the concepts of **continuous** and **differentiable** related?
- Some ways a function can fail to be differentiable at a point.
- Higher derivatives

Derivatives and Rates of Change

Remember the main motivation for introducing limits?

We discovered we need limits when we tried to solve **tangent** and **velocity** problems.

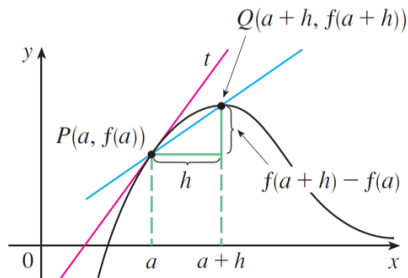
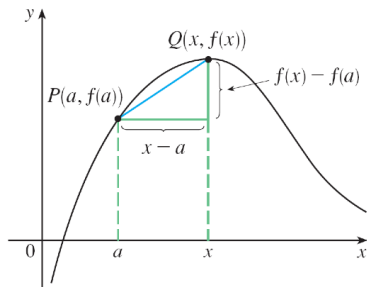
Definition

Let $D \subset \mathbb{R}$ and let $f : D \rightarrow \mathbb{R}$ be a function. Then we define the **tangent line** to the graph of $f(x)$ at the point a to be the line going through the point $(a, f(a))$ with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

if the limit exists. If this limit does not exist then we say that the graph does not have a tangent at that point.

Derivatives and Rates of Change



Instead of working with **the position x** of the second point (left figure), we could equally well work with **the displacement h** from a to that point (right figure). Then the slope of the tangent line becomes

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivatives and Rates of Change

Definition

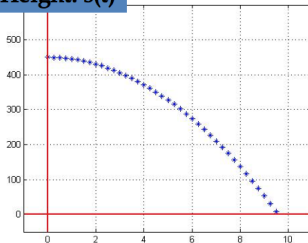
Assume that a particle's trajectory (in one dimension) is described by a function $s : \mathbb{R} \rightarrow \mathbb{R}$. The **instantaneous velocity** of the particle at time t is defined to be:

$$\lim_{\tau \rightarrow 0} \frac{s(t + \tau) - s(t)}{\tau}.$$

Instantaneous velocity:



Height: $s(t)$

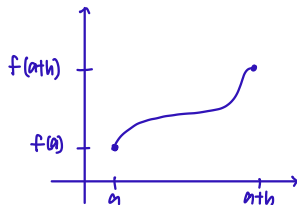


Time

Derivative of a function

Since this type of limit

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



actually occurs widely and arises whenever we calculate a rate of change in any of the sciences or engineering, it is given a special name and notation.

Definition: The derivative at a

The derivative of a function f at a number a , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Derivative of a function

If we write $x = a + h$, then we have $h = x - a$ and h approaches 0 if and only if x approaches a . Therefore an equivalent way of stating the definition of the derivative, as we saw in finding tangent lines, is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

We can now say that

The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a .

Suppose y is a quantity that depends on another quantity x . Thus y is a function of x and we write $y = f(x)$. We can have another interpretation of $f'(a)$:

The derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

Derivative of a function

Exercise

Use the definition of derivative to determine the derivative of the function $f(x) = x^3 - 3x + 2$ at the number a .

$$\begin{aligned} & \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{x^3 - 3x + 2 - (a^3 - 3a + 2)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{(x-a)(x^2 + ax + a^2) - 3(x-a)}{x-a} \\ &= \lim_{x \rightarrow a} (x^2 + ax + a^2 - 3) \end{aligned}$$

by limit laws $\rightarrow$ $= 3a^2 - 3$

can cancel of f
 $x-a \because x \rightarrow a$, it is approaching a

or

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

We were asked to use definition. This means we have to determine if the corresponding limit exists or not.

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(a+h)^3 - 3(a+h) + 2] - (a^3 - 3a + 2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^3 + 3a^2h + 3ah^2 + h^3 - 3a - 3h + 2 - (a^3 - 3a + 2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3a^2h + 3ah^2 + h^3 - 3h}{h} \\ &= \lim_{h \rightarrow 0} (3a^2 + 3ah + h^2 - 3) \\ &= 3a^2 - 3 \end{aligned}$$

The derivative as a function

We have considered the derivative of a function f at a fixed number a

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

Here we let the number a vary and be replaced by a variable x , we obtain the definition of the derivation of f .

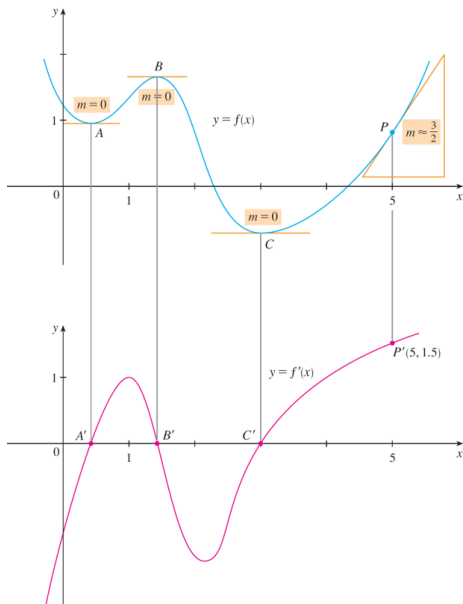
Definition: The derivative function

Let f be function. The **derivative function** of $f(x)$ is defined by the following rule:

$$f'(x) = \begin{cases} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, & \text{if this limit exists at } x \\ \text{undefined,} & \text{otherwise} \end{cases}$$

The domain of $f'(x)$ is the set $\{x | f'(x) \text{ exists}\}$ and **may be smaller than the domain of f .**

Graphs of a function and its derivative



The derivative as a function

Example 3

If $f(x) = \sqrt{x}$, find the derivative of f . State the domain of f'

Solution

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \right) \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}. \end{aligned}$$

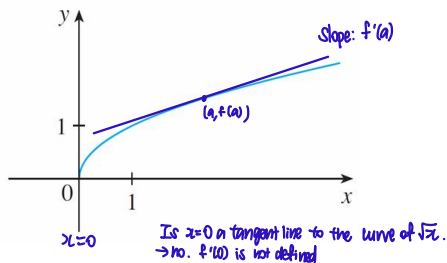
Handwritten notes:

- $\rightarrow$ if $x=0$, $\sqrt{h} \Rightarrow$ not valid if approaching h from left of 0!
- Rationalize the numerator
- $h \neq 0$ so can cancel out h
- So we don't include 0

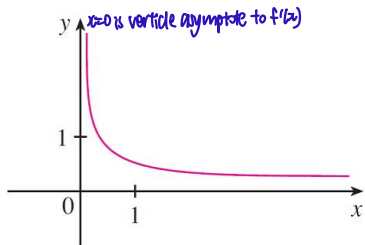
We see that $f'(x)$ exists if $x > 0$, so the domain of f' is $(0, \infty)$. This is slightly smaller than the domain of f , which is $[0, \infty)$.

The derivative as a function

The graph of $f(x) = \sqrt{x}$.



The graph of $f'(x) = 1/(2\sqrt{x})$.



Other notations for the derivative

$y = f(x)$ indicates that the independent variable is x and the dependent variable is y . Some common alternative notations for the derivative are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$$

The symbols D and d/dx are called **differentiation operators** because they indicate the operation of **differentiation**, which is the process of calculating a derivative.

The symbol dy/dx , which was introduced by Leibniz, should not be regarded as a ratio (for the time being); it is simply a synonym for $f'(x)$. We can rewrite the definition of derivative in Leibniz notation in the form

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1} \quad x_2 \neq x_1$$

$\downarrow$
 $x_2 - x_1$

The notation

$$f'(a) = \left. \frac{dy}{dx} \right|_{x=a}$$

is a synonym for $f'(a)$. The vertical bar means *evaluate at*.

Differentiability

Definition

A function $f(x)$ is **differentiable at a** if $f'(a)$ exists. It is **differentiable on an open interval** (a, b) [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Both continuity and differentiability are desirable properties for a function to have. The following theorem shows how these properties are related.

Theorem 4

If $f(x)$ is differentiable at a , then f is continuous at a . $\rightarrow$ then f has a limit at a

Note: The converse of Theorem 4 is false; that is, there are functions that are continuous but not differentiable. We will show one example after proving Theorem 4.

Differentiability

Proof of Theorem 4

Since f is differentiable at a , we know that the following limit exists and equals $f'(a)$:

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

To prove that f is continuous at a , we check the value of $\lim_{x \rightarrow a} f(x)$.

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [(f(x) - f(a)) + f(a)] = \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} (x - a) + f(a) \right]$$

Both $\lim_{x \rightarrow a} (x - a)$ *a polynomial* and $\lim_{x \rightarrow a} f(a)$ *a constant* exist. We can write

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) + \lim_{x \rightarrow a} f(a) \\ &= f'(a) \cdot 0 + f(a) = f(a). \end{aligned}$$

Therefore, f is continuous at a .

The derivative as a function

Example 5

Determine the derivative of $f(x) = |x|$. State the domain of f'

continuous at every point of its domain

Solution

To determine the derivative function we need to determine the derivative limit at every point a of $\mathbb{R}$.

We do this by grouping the points into cases where the calculation is the same.

Case 1: $x > 0$

If $x > 0$, then $|x| = x$ and we can choose h small enough that $x + h > 0$ and hence $|x + h| = x + h$. Therefore, for $x > 0$, we have

*removes possibility that $h > x$,
so $x+h$ will always be > 0 .
 $\Rightarrow h$ can take negative values*

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} 1 = 1. \end{aligned}$$

Example 5–Solution (Cont'd)

Case 2: $x < 0$

h can take positive values

If $x < 0$, then $|x| = -x$ and we can choose h small enough that $x + h < 0$ and hence $|x + h| = -(x + h)$. Therefore, for $x < 0$, we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} \\ &= \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} = \lim_{h \rightarrow 0} (-1) = -1. \end{aligned}$$

Case 3: $x = 0$

For $x = 0$ we have to investigate the existence of $f'(0)$:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}.$$

Example 5–Solution (Cont'd)

Let's compute the left and right limits separately:

$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} (-1) = -1$$

and

$$\lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} 1 = 1.$$

Since these limits are different, $f'(0)$ does not exist. Thus f is differentiable at all x except 0.

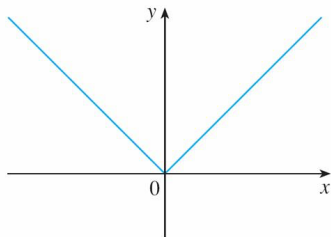
not differentiable at $x=0$

A formula for $f'(x)$ is given by

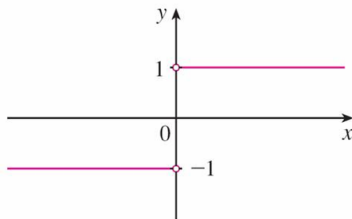
$$f'(x) = \begin{cases} -1, & \text{if } x < 0 \\ 1, & \text{if } x > 0 \end{cases}$$

Example 5–Solution (Cont'd)

The fact that $f'(0)$ does not exist is reflected geometrically in the fact that the curve $y = |x|$ does not have a tangent line at $(0,0)$.



(a) $y = f(x) = |x|$



(b) $y = f'(x)$

Note: $y = f(x)$ is continuous everywhere, but it is not differentiable at $x = 0$.

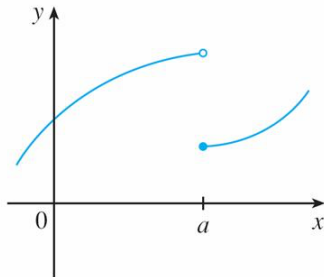
Differentiability

Question. How can a function fail to be differentiable at some a ?

Answer. There are mainly 3 ways that a function can fail to be differentiable at some a .

Possibility 1: f is not continuous at a .

Theorem 4 says that a function is differentiable at a , then it is continuous at a . Equivalently, it says that if f is not continuous at a , then f is not differentiable at a . So at **any discontinuity f fails to be differentiable.**



Differentiability

Possibility 2: f has a **corner** at a . $\rightarrow$ e.g. $|x|$
 $\rightarrow$ it doesn't change smoothly $\rightarrow$ like a curve

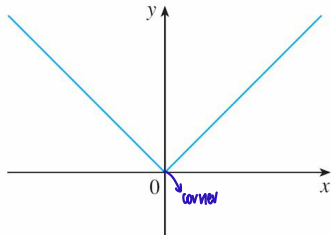
This means that

$$\lim_{h \rightarrow 0^-} \frac{f(x+a) - f(a)}{h}$$

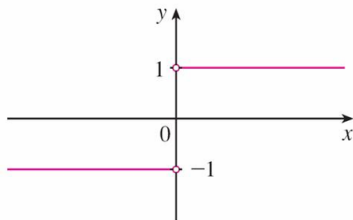
and

$$\lim_{h \rightarrow 0^+} \frac{f(x+a) - f(a)}{h}$$

both exist, but they are **different**.



(a) $y = f(x) = |x|$



(b) $y = f'(x)$

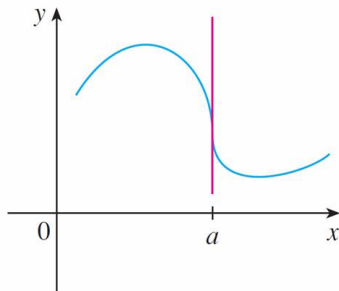
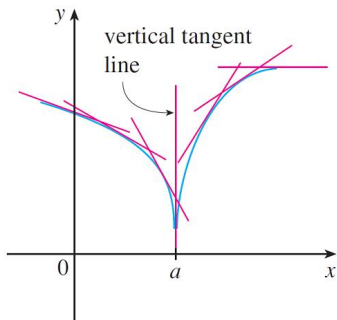
Differentiability

Possibility 3: f has a **vertical tangent** at $a \rightarrow$ e.g. $\sqrt{x}$

This means that *Not vertical line!*

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \infty \quad \text{or} \quad -\infty,$$

indicating that the tangent lines become steeper and steeper as $x \rightarrow a$.



Differentiability

Exercise

Show that the following function f has a vertical tangent at $x = 0$.

$$f(x) = \begin{cases} \sqrt{x}, & \text{if } x \geq 0 \\ -\sqrt{-x}, & \text{if } x < 0 \end{cases}$$

We can determine the quotient at $a=0$ as

$$\left. \frac{f(a+h) - f(a)}{h} \right|_{a=0} = \frac{f(h) - f(0)}{h} = \begin{cases} \frac{1}{\sqrt{h}}, & \text{if } h > 0 \\ \frac{1}{-\sqrt{-h}}, & \text{if } h < 0 \end{cases}$$

Handwritten notes: Red arrows point from the cases to $+\infty, h \rightarrow 0^+$ and $+\infty, h \rightarrow 0^-$. A purple arrow points from the denominator of the second case to $\frac{-f-h-0}{h}$.

Therefore, at $a=0$, we have

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \infty$$

Higher derivatives

If f is a differentiable function, then its derivative f' is also a function, so f' may have a derivative of its own, denoted by $(f')' = f''$. This new function f'' is called the **second derivative** of f because it is the derivative of the derivative of f .

Using Leibniz notation, we write the second derivative of $y = f(x)$ as

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2 y}{dx^2}$$

Example 6

If $f(x) = x^3 - x$, find and interpret $f''(x)$.

Example 6–Solution

Solution

The first derivative of $f(x) = x^3 - x$ is

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[(x+h)^3 - (x+h)] - (x^3 - x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x^3 + 3x^2h + 3xh^2 + h^3 - x - h) - (x^3 - x)}{h} \\&= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 1) = 3x^2 - 1.\end{aligned}$$

Then the second derivative of f is

$$\begin{aligned}f''(x) &= \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 1] - (3x^2 - 1)}{h} \\&= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 1) - (3x^2 - 1)}{h} \\&= \lim_{h \rightarrow 0} (6x + 3h) = 6x.\end{aligned}$$

Acceleration

In general, we can interpret a second derivative as a rate of change of a rate of change. The most familiar example of this is **acceleration**, which we define as follows.

If $s = s(t)$ is the **position function of an object** that moves in a straight line, we know that its first derivative represents the velocity $v(t)$ of the object as a function of time:



$$v(t) = s'(t) = \frac{ds}{dt}.$$

The instantaneous rate of change of velocity with respect to time is called the **acceleration** $a(t)$ of the object. Thus the acceleration function is the derivative of the velocity function and is therefore the second derivative of the position function:

$$a(t) = v'(t) = s''(t)$$

or, in Leibniz notation

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

Higher derivatives

The third derivative f''' is the derivative of the second derivative:

$$f''' = (f'')'.$$

Likewise, $f'''(x)$ can be interpreted as the slope of the curve $y = f''(x)$ or as the rate of change of $f''(x)$.

If $y = f(x)$, then alternative notations for the third derivative are

$$y''' = f'''(x) = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{d^3 y}{dx^3}.$$

The differentiation process can be continued. The fourth derivative f'''' is usually denoted by $f^{(4)}(x)$.

In general, the n -th derivative of f is denoted by $f^{(n)}$ and is obtained from f by differentiating n times. If $y = f(x)$, we write

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n}.$$